

PS 3: Week 5

Randomized Experiments vs. Observational Data

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Week 5

Agenda

- ① Attendance Survey
- ② Populism: Potential Outcomes
- ③ Populism: Observational Data
- ④ Populism: An Experiment
- ⑤ Practice Qs

A Motivating Case: Tylenol & Autism

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- **Is this causal?**

A Motivating Case: Tylenol & Autism

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- **Is this causal?**
- **Omitted variable bias:** people who take Tylenol may do so because of other conditions (chronic pain, environmental factors, etc.) that could *also* raise autism risk.
- **Selection bias:** people who choose to take a “treatment” may differ, in ways connected to the outcome, from those who don’t.

A Potential Causal Effect: Populism

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- Campaigns want to know whether **populist rhetoric** – framing some “elite” (a party, billionaires, cultural elites) as unfairly ruling over “the people” – actually persuades voters.
- Imagine voters in a 2026 campaign: some are exposed to populist rhetoric, and each later votes for Party P or Party O.

Discuss:

- What's the **treatment** in this example?
- What are the **potential outcomes**?
- What is each individual's **treatment effect**?

Potential Outcomes

Voter	Vote if exposed?	Vote if not exposed?	Individual Treatment Effect
Tess	0	0	0
Jeremy	1	1	0
Anne Marie	1	0	1
Jack	0	1	-1

1: *Party P*

0: *Party O*

Observational Data on Populism

- A survey of over 1 million voters asks: were you exposed to Party P's populist rhetoric in 2026 (TV, social media, mailers, etc.)? Who did you vote for?
- People exposed to the rhetoric are **far more likely** to vote for Party P.

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Discuss:

- How confident should we be that this is *causal*?
- What sources of bias might undermine this finding?
- What are the *advantages* of a dataset like this?

Omitted Variable Bias, Selection Bias, and Reverse Causation!

Voter	Exposed in 2025?	Voted P 2025?	Voted P 2021?
Tess	1	1	1
Jeremy	1	1	1
Anne Marie	0	0	0
Jack	0	0	0

How Do We Resolve These Biases?

- With observational data, people can **select into treatment**.
- If that selection *and* the outcome are both explained by some other variable, that's **omitted variable bias**.
 - E.g., men being less risk-averse → more likely to take an experimental drug *and* have lower resting heart rates.
- **Randomization** makes treatment and control groups similar on *all* variables (given a large enough sample) – no more omitted variable bias to worry about.
 - E.g., prior vote can no longer cause *both* exposure to rhetoric and future vote, because it's balanced across groups.

How might we set this up?

- What's the treatment?
- How is the treatment assigned?
- What outcome(s) are we measuring?
- What variables should we measure?
 - Often driven by which **omitted variables** we're worried about.

Bundled Treatments

- To find a causal effect, you need to **isolate** your explanatory variable from your outcome.
- If your treatment actually bundles *multiple* things together, how do you know which part is doing the work?
 - Is it the **populism**? Or the **policies**?
- Solution: split the bundle across multiple treatment arms.
 - Populism treatment (previous slide)
 - Policies only, no populist framing
 - Control group
- Compare the bundled treatment to the “policies only” arm to isolate the effect of populist rhetoric *itself*.

Practice Coding Exercise 1

- Note: *This is a harder exercise than we have done in class or in the problem set before.*

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- Using the data set on persuasion and voting from Wednesday in `Week5_Sandbox1.ipynb`
- What is the turnout percentage for each treatment? (Civic Duty, Hawthorne, Self, and Neighbors)

Answer: Coding Exercise 1

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We could also do it all in one step:

Practice Coding Exercise 2: Random Assignment

In the in-class activity we checked if age differed between the Neighbors and Control groups.

Does voter-history (g2000, g2002, p2004) differ by treatment assignment?

That is, what is the turnout rate for each election across the 4 treatments and control group?

What does this tell us about random assignment?

Practice Coding Exercise 2: Answers

Remember how we subset the data before:

Why did we need to do this for this problem as well?

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Remember how we subset the data before:

Why did we need to do this for this problem as well?

Because we wanted to know treatment assignment across several groups.

What does this mean?

What does this mean?

There are similar values across each condition which means random assignment of treatment likely worked

g2002 Random Assignment

p2004 Random Assignment

Acknowledgements

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